Continuous Optimization for Airline Hub-and-Spoke Network Design

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Abstract—**We study the design of airline hub-and-spoke networks through a continuous optimization framework that jointly models network deployment, passenger choice, and market profitability. The lower-level model combines operating costs, sparse deployment penalties, price-weighted passenger utility terms, and entropy regularizers that recover a logit demand split between the new airline and the outside alternative. This yields a continuous approximation of hub activation and route opening decisions while preserving the economic structure of the airline design problem.**

On top of this model, we introduce a bilevel formulation in which market weights are learned rather than fixed. The upper level maximizes the real objective of the airline, namely profit, by adjusting market-specific hyperparameters that scale the lower-level utility and entropy terms. This allows the model to emphasize the origin-destination markets that are most relevant for profitability, instead of relying on exogenous weights chosen *a priori*.

The main difficulty is the simultaneous presence of hierarchical optimization and fixed deployment costs. To address it, we combine a reweighted ℓ_1 approximation of fixed costs with a penalty-based bilevel strategy. The resulting algorithm alternates an outer sparsity loop, which updates the reweighting coefficients associated with capacities, and an inner penalty loop, which alternates between a penalized bilevel solve and a lower-level solve to update the market hyperparameters. The final methodology provides a practical route to solving airline network design problems that are beyond the reach of exact mixed-integer bilevel formulations, while maintaining an explicit link between passenger behavior, network topology, and profit.

I. Introduction

Transportation networks are a critical component of modern societies. They support economic activity, territorial cohesion, access to employment and services, and the movement of passengers and goods at multiple spatial and temporal scales. However, the goals of network design depend strongly on the application. In road networks, the emphasis is typically placed on reducing travel times and mitigating congestion. In rapid transit and public transport systems, planners are often more concerned with coverage, accessibility, and social cohesion. In contrast, privately operated transportation systems are mainly designed from a profitability viewpoint, where the network configuration must balance service attractiveness against operating and infrastructure costs. From an optimization perspective, these settings all lead to network design problems, but with markedly different objective functions and modeling choices.

Within this broad class of problems, hub-and-spoke networks play a particularly important role in air transportation. Structurally, they can be interpreted as the union of several star-shaped subnetworks interconnected through a reduced set of hub airports. Spoke airports mainly feed or receive local traffic, whereas hubs concentrate transfer activity and allow the operator to connect many origin-destination markets with a comparatively limited number of direct links. For airlines, this architecture is attractive because it consolidates traffic, improves the use of aircraft and airport resources, and allows broad market coverage without operating every origin-destination pair as a direct service. Consequently, hub-and-spoke network design for a private airline is fundamentally a profit-driven problem in which the key decisions concern hub activation and the assignment of capacities across airports and routes. Passenger flow variables are incorporated not as direct operational decisions, but as a modeling device to represent how users may react to the offered network and, therefore, how much revenue each market can generate for the operator.

Classical formulations address these decisions through mixed-integer programming (MIP) models, since fixed-cost activation decisions are naturally represented with binary variables. This is the standard modeling approach for hub location and network design, but the resulting formulations become combinatorially challenging as the number of airports, markets, and operating restrictions grows, with complexity increasing very rapidly once multicommodity interactions and fixed costs are included. Modeling passenger behavior creates an additional challenge. Simplified all-or-nothing assignment rules are linear and easy to embed in optimization models, but they provide a poor representation of how users choose among competing itineraries. Logit-based demand models are considerably more realistic because they allow the captured share of each market to vary smoothly with the perceived utility of the available alternatives. Yet, when introduced into a network design problem, they typically give rise to nonlinear and nonconvex constraints, which are often handled through piecewise-linear mixed-integer approximations or, alternatively, through exponential-cone formulations.

This paper builds on a continuous optimization framework in which reweighted ℓ_1 sparsity is used to approximate the fixed costs associated with airport, hub, and route activation, while entropy-based regularization recovers logit market shares without explicitly imposing nonconvex share equations. Those ingredients provide the lower-level continuous model on top of which we develop the main contribution of this work: a bilevel formulation with adjustable market-specific coefficients that fine-tune the lower-level objective so that the resulting network better reflects the airline's actual profit criterion. To solve the resulting problem, we adopt a penalty-based bilevel reformulation combined with an outer iteration that updates the reweighted ℓ_1 terms, thereby improving the approximation of real fixed costs across successive solves.

The remainder of the paper is organized as follows. We first review the relevant literature on hub-and-spoke network design, continuous approximations of fixed-cost models, and demand modeling. We then describe the airline problem setting and present the continuous lower-level formulation. Next, we introduce the bilevel reformulation and the associated solution strategy. Finally, we report computational experience on benchmark and larger-scale instances, and close with the main conclusions.

II. State of the Art

a) Classical hub-and-spoke network design.: Hub-and-spoke network design has traditionally been modeled as a discrete location-and-routing problem in which the main decisions are the selection of hub facilities, the assignment of spoke nodes, and the activation of direct or inter-hub services. Foundational formulations cast these decisions as mixed-integer optimization models, often under single-allocation or multiple-allocation rules and with complete or incomplete hub-level subnetworks [1]. Over time, this line of work has expanded toward richer network-design variants that incorporate capacities, incomplete hub networks, fixed charges, and operational restrictions, while preserving the view that hub activation and service design are fundamentally combinatorial decisions [2], [3], [4]. In airline applications, this family of models remains attractive because it mirrors the binary nature of opening facilities and operating services, but the resulting formulations become difficult to scale once multicommodity flows, route feasibility, and market-level economics are represented explicitly.

b) Airline demand modeling and route choice.: A second line of work concerns how demand is represented within network design models. In simplified formulations, origin–destination demand is often treated as fixed and is routed deterministically through the designed network. This approximation is computationally convenient, but it suppresses the fact that passengers react to fares, travel times, and service quality. Discrete-choice models, and in particular multinomial logit specifications, provide a much more behaviorally meaningful description of demand capture in transportation systems [5]. However, once market shares depend endogenously on network design decisions, the optimization model inherits nonlinear and frequently nonconvex couplings between design, utility, and demand allocation. For airline hub-and-spoke design, this is precisely the point at which realistic demand modeling begins to conflict with tractable large-scale optimization.

c) Continuous and sparsity-based approximations.: These computational difficulties have motivated a different modeling philosophy in which some binary activation decisions are replaced by sparse continuous variables. Instead of deciding directly whether an airport, hub, or service is open, one may optimize continuous capacities and then promote sparse solutions so that inactive components collapse naturally to zero. In this context, reweighted ℓ_1 schemes derived from nonconvex sparsity surrogates have become an influential tool for approximating fixed-cost activation patterns while retaining a continuous optimization structure [6]. The same philosophy can be extended to the demand side: rather than imposing explicit logit equations, market splitting can be induced through smooth entropy regularization, which preserves convexity and yields endogenous shares through optimality conditions. Although these ideas are well established in continuous optimization, they are still much less common in airline hub-and-spoke network design than the classical mixed-integer viewpoint.

d) Bilevel optimization and penalty methods.: Bilevel optimization provides a natural framework when one layer of decisions is used to calibrate or influence the solution of another. The field has developed from early Stackelberg and penalty-based formulations [7], [8] into a broad methodological literature covering exact reformulations, descent methods, evolutionary approaches, and modern first-order algorithms [9], [10], [11]. For the class of models most relevant here, namely problems in which the lower level remains a structured continuous optimization problem, penalty and value-function ideas are especially appealing because they avoid an explicit enumeration of lower-level optimality conditions inside a large discrete model. Recent first-order penalty methods have reinforced this viewpoint by showing how constrained bilevel problems with convex lower-level structure can be approached through penalized single-level surrogates with implementable complexity guarantees [12].

e) Position of the present work.: The present paper lies at the intersection of these four strands. Relative to the traditional hub-location literature, it avoids an explicit mixed-integer treatment of airport, hub, and service activation and instead works with a continuous sparse design model. Relative to standard deterministic network-design formulations, it incorporates endogenous demand capture through utility and entropy terms so that market shares react to the designed network. Relative to generic sparsity-based continuous models, it is tailored to airline hub-and-spoke design and to a profit-oriented private operator. Finally, relative to the general bilevel literature, the upper level is not used here to redesign the network directly, but to calibrate market-specific coefficients that guide the lower-level continuous approximation toward the airline's true objective. This combination of entropy-based demand modeling, sparse continuous activation, and bilevel calibration is, to our knowledge, not standard in the airline hub-and-spoke design literature and defines the methodological contribution developed in the next sections.

III. Problem Setting

We consider the design of a hub-and-spoke air transportation network for a private airline entering a competitive market. Let $\mathcal{S}$ denote the set of candidate airports, $\mathcal{A} \subseteq \mathcal{S} \times \mathcal{S}$ the set of candidate flight legs, and $\mathcal{K}$ the set of origin-destination markets with positive demand. For each market $(o, d) \in \mathcal{K}$, the entrant competes against an outside alternative representing incumbent airlines and, when appropriate, the no-travel option.

The airline operates under a hub-and-spoke architecture. Some airports may be activated only as spokes, mainly processing local departures and arrivals, whereas others may also receive additional transfer capacity and operate as hubs. This distinction is central in the airline setting because local airport activity and connecting activity generate different resource requirements. As a result, the design problem must decide not only which airports are active, but also which of them are upgraded to hubs, how much spoke and hub capacity is installed at each airport, and what service capacity is assigned to each candidate connection.

Passenger behavior is endogenous. For every market (o, d), the model determines the demand share captured by the entrant and the itinerary followed by that demand through the designed network. These flow variables are not interpreted as direct operational controls; rather, they are introduced to approximate how passengers react to the offered service and, therefore, how much revenue the network can generate. This is the mechanism through which the optimization model internalizes the interaction between network structure and market attractiveness.

A. Sets, Parameters, and Variables

To state the model precisely, we use the following notation.

Sets:

- $\mathcal{S}$ is the set of candidate airports.
- $\mathcal{A} \subseteq \mathcal{S} \times \mathcal{S}$ is the set of candidate flight legs.
- $\mathcal{K}$ is the set of origin-destination markets with positive demand.
- $\mathcal{Q}^{od}_{\text{ext}}$ is the set of outside alternatives available in market (o, d).
- $\mathcal{N}_{\text{in}}(i)$ and $\mathcal{N}_{\text{out}}(i)$ denote the incoming and outgoing neighbors of airport i.

Parameters:

- w^{od} is the demand in market (o, d) and p^{od} is the fare charged by the entrant in that market.
- v^{od}_q is the utility of outside alternative $q \in \mathcal{Q}^{od}_{\text{ext}}$.
- t_{ij} is the travel time associated with leg (i, j).
- c_{s_i} and c_{h_i} are the fixed costs of activating airport i as a spoke and as a hub, respectively.
- $\bar{c}_{s_i}$ is the variable cost of installing airport capacity at node i, and $c_{o_{ij}}$ is the operating cost of leg (i, j).
- a_{nom} is the nominal aircraft capacity, τ is the route load factor, and σ and σ_i are scaling coefficients for transfer and airport processing capacity.
- λ weights hub fixed costs inside the budget, $a_{\max}$ is the total fleet limit, b is the available budget, and M is a sufficiently large constant.
- ω_p and ω_t are the coefficients associated with fare and travel time in the passenger utility model.

Decision variables:

- $S^{\mathfrak{b}}_i \in \{0, 1\}$ indicates whether airport i is active, and $S^{h\mathfrak{b}}_i \in \{0, 1\}$ indicates whether airport i operates as a hub.
- $S_i \geq 0$ and $S^h_i \geq 0$ are the spoke and hub capacities installed at airport i.
- $A_{ij} \geq 0$ is the service capacity assigned to leg (i, j).
- $F^{od} \in [0, 1]$ is the share of market (o, d) captured by the entrant.
- F^{od}_{ij} indicates whether the itinerary used by market (o, d) traverses leg (i, j).
- Z^{od} indicates whether market (o, d) is served by the entrant.

For compactness, the design variables are grouped as

$$\mathcal{X}_{\text{ser}} = \{S^{\mathfrak{b}}_i, S^{h\mathfrak{b}}_i, S_i, S^h_i, A_{ij}\}, \tag{1}$$

whereas the passenger and routing variables are grouped as

$$\mathcal{X}_{\text{pax}} = \{F^{od}, F^{od}_{ij}, Z^{od}\}. \tag{2}$$

B. Mixed-Integer Viewpoint

To motivate the continuous reformulation developed later, we first describe the underlying mixed-integer viewpoint. In this section, capital letters denote the classical design variables: S_i and S^h_i are the spoke and hub capacities installed at airport i, A_{ij} is the service capacity assigned to leg (i, j), $S^{\mathfrak{b}}_i$ indicates whether airport i is active, and $S^{h\mathfrak{b}}_i$ indicates whether airport i operates as a hub. For passenger assignment, F^{od} denotes the share of market (o, d) captured by the entrant, and F^{od}_{ij} indicates whether the itinerary used by market (o, d) traverses leg (i, j).

Since the operator is private, the natural objective is profit maximization. Equivalently, the model can be written as the minimization of negative profit:

$$\min \; -\sum_{(o,d)\in\mathcal{K}} w^{od} p^{od} F^{od} + \sum_{(i,j)\in\mathcal{A}} c_{o_{ij}} A_{ij}, \tag{3}$$

where the first term represents passenger revenue and the second term represents the operating cost of the flights offered by the entrant. The design therefore balances market capture against the cost of supplying the corresponding air service.

C. Constraint Families

The mixed-integer viewpoint combines infrastructure activation decisions, service-capacity constraints, passenger-routing constraints, and demand-capture constraints. For clarity, we discuss these restrictions by family.

1) Airport Activation and Budget Constraints: The hub-and-spoke structure requires two airport-capacity variables. Spoke capacity S_i accounts for local departures and arrivals, whereas hub capacity S^h_i accounts for transfer activity. Their binary activation variables satisfy

$$S_i \leq M S^{\mathfrak{b}}_i \qquad \forall i \in \mathcal{S}, \tag{4}$$

$$S^h_i \leq M S^{h\mathfrak{b}}_i \qquad \forall i \in \mathcal{S}, \tag{5}$$

$$S^{h\mathfrak{b}}_i \leq S^{\mathfrak{b}}_i \qquad \forall i \in \mathcal{S}, \tag{6}$$

Constraint (4) activates spoke capacity only when airport i is opened, and (5) plays the same role for hub capacity. Constraint (6) enforces the hierarchical logic of the design: an airport can operate as a hub only if it is already active in the network. In contrast, we do not impose a big-M activation relation on route capacities, because the current airline formulation does not associate fixed costs with arcs.

The infrastructure budget must account for spoke activation, hub activation, and installed airport capacity. This leads to

$$\sum_{i\in\mathcal{S}} c_{s_i} S_i^{\mathfrak{b}} + \lambda \sum_{i\in\mathcal{S}} c_{h_i} S_i^{h\mathfrak{b}} + \sum_{i\in\mathcal{S}} \bar{c}_{s_i}\big(S_i + S_i^h\big) \le b, \qquad (7)$$

where λ modulates the relative cost of opening hubs. This restriction captures the strategic trade-off between enlarging the airport system and preserving enough resources to support flight operations. The total number of flights is also limited by the available fleet:

$$\sum_{(i,j)\in\mathcal{A}} A_{ij} \le a_{\max}. \qquad (8)$$

Constraint (8) prevents the model from assigning more service capacity than can be supported by the available aircraft. Because aircraft are assumed to overnight at hubs, the service pattern must also be symmetric, which yields

$$A_{ij} = A_{ji} \qquad \forall (i,j) \in \mathcal{A}. \qquad (9)$$

This symmetry condition is a stylized representation of the hub-and-spoke operating pattern and avoids one-way capacity imbalances that would be unrealistic in the strategic design stage.

2) Leg and Airport Capacity Constraints: Service capacity, airport capacity, and transfer capacity are jointly constrained by the hub-and-spoke logic. The capacity of each leg must be sufficient to transport the demand assigned to it:

$$\sum_{(o,d)\in\mathcal{K}} w^{od} F^{od} F_{ij}^{od} \le a_{\text{nom}} \tau A_{ij} \qquad \forall (i,j) \in \mathcal{A}, \qquad (10)$$

where a_{nom} is the nominal aircraft capacity and τ is the load factor. Constraint (10) therefore ensures that the aggregate number of passengers routed through leg (i,j) does not exceed the effective transport capacity supplied on that leg.

Local airport capacity is enforced through the aggregate arrival and departure workload,

$$\sigma_i \sum_{j\in\mathcal{N}_{\text{in}}(i)} A_{ji} + \sigma_i \sum_{j\in\mathcal{N}_{\text{out}}(i)} A_{ij} \le S_i + S_i^h \qquad \forall i \in \mathcal{S}, \quad (11)$$

which states that the total local workload induced by incoming and outgoing flights must be supported by the installed airport capacity. Since hubs process not only local traffic but also connecting passengers, transfer activity is limited separately through

$$\sigma\Bigg(\sum_{j\in\mathcal{N}_{\text{out}}(i)} \sum_{\substack{(o,d)\in\mathcal{K}\\ o\neq i}} \frac{w^{od} F^{od} F_{ij}^{od}}{a_{\text{nom}}\tau} + \sum_{j\in\mathcal{N}_{\text{in}}(i)} \sum_{\substack{(o,d)\in\mathcal{K}\\ d\neq i}} \frac{w^{od} F^{od} F_{ji}^{od}}{a_{\text{nom}}\tau}\Bigg) \le S_i^h \qquad \forall i \in \mathcal{S}. \quad (12)$$

Constraint (12) captures the fact that connecting passengers can only be processed if the corresponding airport has enough hub capacity. This is one of the defining structural features of the model, because it distinguishes the role of a spoke from the role of a hub. In addition, a direct service between the endpoints of a market is allowed only if at least one of those endpoints is operated as a hub:

$$\sigma \frac{w^{od} F^{od} F_{od}^{od}}{a_{\text{nom}}\tau} \le S_o^h + S_d^h \qquad \forall (o,d) \in \mathcal{K}. \qquad (13)$$

Constraint (13) prevents the model from generating direct services that are incompatible with the hub-and-spoke architecture.

3) Passenger Routing and Market Feasibility Constraints: The passenger flow variables must satisfy the usual conservation equations. If Z^{od} denotes whether market (o,d) is served by the entrant, then

$$\sum_{j\in\mathcal{N}_{\text{out}}(i)} F_{ij}^{od} - \sum_{j\in\mathcal{N}_{\text{in}}(i)} F_{ji}^{od} = \begin{cases} Z^{od}, & i = o, \\ -Z^{od}, & i = d, \\ 0, & \text{otherwise}, \end{cases} \qquad (14)$$

for every $i \in \mathcal{S}$ and $(o,d) \in \mathcal{K}$. Constraint (14) ensures that each served market has a valid path from its origin to its destination and that no artificial creation or destruction of flow occurs at intermediate nodes.

To avoid attracting markets that are not profitable, the revenue collected on each served market must dominate the operating cost induced by its itinerary:

$$w^{od} p^{od} F^{od} \ge \sum_{(i,j)\in\mathcal{A}} \frac{w^{od} c_{o_{ij}} F^{od} F_{ij}^{od}}{a_{\text{nom}}\tau} \qquad \forall (o,d) \in \mathcal{K}. \quad (15)$$

Constraint (15) filters out unattractive itineraries by requiring each captured market to be self-supporting from the airline's viewpoint.

4) Demand-Capture Constraint: Passenger choice is represented through a logit-type capture rule. The utility of the entrant in market (o,d) combines the fare charged by the airline and the total travel time of the selected itinerary, namely

$$u_{\text{self}}^{od} = \omega_p p^{od} + \omega_t \sum_{(i,j)\in\mathcal{A}} t_{ij} F_{ij}^{od}. \qquad (16)$$

The captured share is then constrained by

$$F^{od} \le \frac{\exp\big(u_{\text{self}}^{od}\big)}{\exp\big(u_{\text{self}}^{od}\big) + \sum_{q\in\mathcal{Q}_{\text{ext}}^{od}} \exp\big(v_q^{od}\big)} \qquad \forall (o,d) \in \mathcal{K}, \qquad (17)$$

Constraint (17) couples network design with passenger behavior in a nonconvex way. The share captured by the entrant depends on the utility generated by the network that has just been designed, while that same utility depends on the routing pattern selected for the market. This feedback is precisely what makes the original mixed-integer formulation difficult to solve at scale. The next section replaces this explicit nonlinear demand-capture equation by an equivalent continuous viewpoint in which fixed costs are approximated through sparsity-inducing penalties and the logit split is recovered through entropy regularization.

IV. Bilevel Continuous Formulation

This section presents the central methodological contribution of the paper. Starting from the mixed-integer nonlinear model introduced in the previous section, we build a continuous approximation that remains faithful to the economics of the airline design problem while being substantially cheaper to solve. On top of that continuous approximation, we introduce a second decision layer that calibrates how each market is weighted in the model so that the resulting network is better aligned with the true objective of the airline, namely profit maximization.

Viewed in simple terms, the proposed method combines two nested tasks. The first task determines a hub-and-spoke network, its capacities, and the market shares induced by that network under a smooth behavioral model. The second task modifies a set of market-specific coefficients so that the first task places the right emphasis on the markets that matter most for profitability. This nested structure is the reason why the model is bilevel.

A. Bilevel Optimization

Bilevel optimization studies problems in which one optimization problem is constrained by the solution set of another. In abstract form, a bilevel problem can be written as

$$\max_{u,x} \quad F(u,x) \tag{18}$$

$$\text{s.t.} \quad x \in \arg\min_{y\in X(u)} f(u,y), \tag{19}$$

where the upper level chooses the variable u while anticipating that the variable x must be an optimizer of the lower-level problem. The main difficulty is that the upper-level feasible set is defined implicitly through the optimality of another optimization problem.

In the present work, the upper-level variables are the market coefficients

$$u := \big(\alpha, \beta\big),$$

whereas the lower-level variables are the continuous design and flow variables of the airline model. The lower level returns the network induced by a given choice of market coefficients, while the upper level chooses those coefficients in order to improve the actual profit of the airline. In this sense, the upper level does not directly build the network. Instead, it learns how the continuous design model should value each market.

The remainder of the section is organized as follows. We first present the continuous design model that is solved inside the nested scheme. We then formulate the corresponding bilevel problem, introduce its value-function penalty reformulation, and explain the admissible set of hyperparameters. Finally, we describe the reweighted ℓ_1 approximation used to represent fixed activation costs in a continuous way.

B. Continuous Design Model

Let

$$X := \big(S, S^h, A, F, F^{\text{ext}}, \{F_{ij}^{od}\}_{(i,j),(o,d)}\big) \tag{20}$$

denote the continuous design variables of the model. The notation respects the variables introduced in the previous section, but the interpretation changes in two important ways. First, binary activation variables are removed and replaced by sparse continuous capacities: positive values of S_i, S_i^h, and A_{ij} indicate activated infrastructure or service, whereas zero values indicate that the corresponding resource is not used. Second, the route variables F_{ij}^{od} are no longer binary itinerary indicators. In the continuous model, they are relaxed to lie in $[0,1]$ and represent the fraction of market (o,d) that is routed through leg (i,j). As a consequence, the binary service indicator Z^{od} of the mixed-integer model is no longer required explicitly.

Each market $(o,d) \in \mathcal{K}$ is assigned two coefficients, α^{od} and β^{od}, which together define the market weight

$$\varphi^{od} := \alpha^{od}\big(p^{od}w^{od}\big)^{\beta^{od}}. \tag{21}$$

These coefficients do not alter the feasible set. Their only function is to modulate the contribution of each market to the continuous objective. In this way, the model can adapt not only to passenger volumes, but also to revenue potential and strategic relevance.

For fixed market coefficients, the continuous model is

$$v(\alpha,\beta) := \min_{X\in\mathcal{X}_c} \Phi_{\mathrm{c}}(X;\alpha,\beta), \tag{22}$$

where $\mathcal{X}_c$ denotes the continuous feasible set and

$$\begin{aligned}\Phi_{\mathrm{c}}(X;\alpha,\beta) := \mathrm{OP}(A) + \mathrm{U}_{\text{self}}(X;\alpha,\beta) + \mathrm{U}_{\text{ext}}(X;\alpha,\beta)\\ + \mathrm{H}_{\text{self}}(X;\alpha,\beta) + \mathrm{H}_{\text{ext}}(X;\alpha,\beta).\end{aligned} \tag{23}$$

The objective is the sum of an operating-cost term, a utility term for the entrant, a utility term for the outside alternative, and two entropy terms that regularize the market split.

a) Operating costs.: The direct flight operating cost is

$$\mathrm{OP}(A) = \sum_{(i,j)\in\mathcal{A}} c_{o_{ij}} A_{ij}. \tag{24}$$

This term penalizes dense or high-capacity route structures and preserves the economic trade-off between market attractiveness and network operating effort.

b) Utility terms.: For market (o, d), the systematic utility of the entrant is represented by

$$u^{od}_{\text{self}}(X) := \omega_p p^{od} + \omega_t \sum_{(i,j)\in\mathcal{A}} t_{ij} F^{od}_{ij}, \tag{25}$$

where ω_p and ω_t measure the sensitivity to fare and travel time. The utility contribution of the entrant in the lower-level objective is then written as

$$\mathrm{U}_{\text{self}}(X;\alpha,\beta) = -\sum_{(o,d)\in\mathcal{K}} \varphi^{od} u^{od}_{\text{self}}(X) F^{od}, \tag{26}$$

while the outside alternative contributes

$$\mathrm{U}_{\text{ext}}(X;\alpha,\beta) = -\sum_{(o,d)\in\mathcal{K}} \varphi^{od} v^{od}_{\text{ext}} F^{\text{ext},od}. \tag{27}$$

Since the lower level is a minimization problem, the negative sign in both terms means that solutions with larger utility are preferred. The market weight φ^{od} determines how strongly that preference influences the resulting network.

c) Entropy regularization and endogenous logit shares.: The continuous formulation also includes

$$\mathrm{H}_{\text{self}}(X;\alpha,\beta) = \sum_{(o,d)\in\mathcal{K}} \varphi^{od} F^{od}\big(\log(F^{od}) - 1\big), \tag{28}$$

$$\mathrm{H}_{\text{ext}}(X;\alpha,\beta) = \sum_{(o,d)\in\mathcal{K}} \varphi^{od} F^{\text{ext},od} \left(\log\left(\frac{F^{\text{ext},od}}{n_{\text{airlines}}}\right) - 1\right). \tag{29}$$

These terms are much more than a smoothing device. Their role is to replace the explicit nonlinear logit equations of the original model by a convex variational representation of market choice. The key idea is that the log-sum-exp structure underlying multinomial logit admits a convex-conjugate representation through negative entropy.

For each market, one may interpret the pair

$$\big(F^{od}, F^{\text{ext},od}\big)$$

as the solution of a variational market-splitting problem over the simplex

$$F^{od} + F^{\text{ext},od} = 1, \qquad F^{od} \geq 0, \qquad F^{\text{ext},od} \geq 0.$$

Within that simplex, the linear utility terms favor the alternative with the highest utility, whereas the entropy terms penalize overly concentrated allocations and create a strictly convex trade-off. The result is a smooth market split that avoids unrealistic all-or-nothing assignments and, at the same time, recovers the same qualitative logic as a logit model.

More formally, the entropy terms appear because the convex conjugate of the log-sum-exp function yields a maximization over the simplex involving linear utilities and negative entropy. After changing signs to fit the minimization structure of the lower level, the model arrives exactly at the combination of utility and entropy terms written above. Therefore, the market shares are not imposed externally. They arise endogenously as optimal solutions of the lower-level problem. This is the reason why the continuous model preserves a behaviorally meaningful demand split while remaining smooth, convex, and computationally tractable.

d) Budget and structural constraints.: The continuous model is not only a change in the objective; it also modifies how feasibility is expressed. The continuous feasible set $\mathcal{X}_c$ is defined by the structural restrictions of the hub-and-spoke problem, but with the binary decisions replaced by sparse capacities and with the passenger-routing variables relaxed to continuous values. More specifically, $\mathcal{X}_c$ is characterized by capacity and symmetry restrictions,

$$\sum_{(o,d)\in\mathcal{K}} w^{od} F^{od}_{ij} \leq a_{\text{nom}} \tau A_{ij} \quad \forall (i,j)\in\mathcal{A}, \tag{30}$$

$$\sigma\left(\sum_j A_{ji} + \sum_j A_{ij}\right) \leq S_i + S^h_i \qquad \forall i \in \mathcal{S}, \tag{31}$$

$$\sigma\left(\sum_{\substack{(o,d),j\\ o\neq i}} \frac{w^{od} F^{od}_{ij}}{a_{\text{nom}}\tau} + \sum_{\substack{(o,d),j\\ d\neq i}} \frac{w^{od} F^{od}_{ji}}{a_{\text{nom}}\tau}\right) \leq S^h_i \qquad \forall i \in \mathcal{S}, \tag{32}$$

$$\sigma \frac{w^{od} F^{od}_{od}}{a_{\text{nom}}\tau} \leq S^h_o + S^h_d \quad \forall (o,d)\in\mathcal{K}, \tag{33}$$

$$A_{ij} = A_{ji} \qquad \forall (i,j)\in\mathcal{A}, \tag{34}$$

$$\sum_{(i,j)\in\mathcal{A}} A_{ij} \leq a_{\max}, \tag{35}$$

flow-conservation and market-split restrictions,

$$\sum_j F^{od}_{oj} - \sum_i F^{od}_{io} = 1 - F^{\text{ext},od} \qquad \forall (o,d)\in\mathcal{K}, \tag{36}$$

$$\sum_j F^{od}_{dj} - \sum_i F^{od}_{id} = -1 + F^{\text{ext},od} \qquad \forall (o,d)\in\mathcal{K}, \tag{37}$$

$$\sum_j F^{od}_{ij} - \sum_j F^{od}_{ji} = 0 \qquad \forall i \notin \{o,d\},\ (o,d)\in\mathcal{K}, \tag{38}$$

$$F^{od} + F^{\text{ext},od} = 1 \qquad \forall (o,d)\in\mathcal{K}, \tag{39}$$

$$w^{od} p^{od} F^{od} \geq \sum_{(i,j)\in\mathcal{A}} \frac{w^{od} c_{o_{ij}} F^{od}_{ij}}{a_{\text{nom}}\tau} \quad \forall (o,d)\in\mathcal{K}, \tag{40}$$

and the reweighted budget and domain restrictions,

$$\sum_{i\in\mathcal{S}} \bar{c}_{s_i}(S_i + S_i^h) + \sum_{i\in\mathcal{S}} \frac{c_{s_i}(S_i + S_i^h)}{\bar{S}_i + \bar{S}_i^h + \varepsilon} + \sum_{i\in\mathcal{S}} \lambda \frac{c_{h_i} S_i^h}{\bar{S}_i^h + \varepsilon} \le b, \tag{41}$$

$$\begin{aligned} 0 \le F_{ij}^{od} &\le 1 && \forall (i,j) \in \mathcal{A},\ (o,d) \in \mathcal{K}, \\ 0 \le F^{od} \le 1, \qquad 0 \le F^{\mathrm{ext},od} &\le 1 && \forall (o,d) \in \mathcal{K}, \\ S_i, S_i^h &\ge 0 && \forall i \in \mathcal{S}, \\ A_{ij} &\ge 0 && \forall (i,j) \in \mathcal{A}. \end{aligned} \tag{42}$$

The main differences with respect to the mixed-integer model are now explicit: binary activation variables disappear, route-use variables become continuous, and the fixed-cost budget is replaced by the reweighted continuous restriction (41).

C. Bilevel Calibration and Penalty Reformulation

Once the continuous design model has been fixed, the outer task is to select the coefficients α^{od} and β^{od} so that the resulting design performs well under the true profit function

$$\Pi(X) = \sum_{(o,d)\in\mathcal{K}} p^{od} w^{od} F^{od} - \sum_{(i,j)\in\mathcal{A}} c_{o_{ij}} A_{ij}. \tag{43}$$

The bilevel problem is therefore

$$\max_{\alpha,\beta,X} \quad \Pi(X) \tag{44}$$

$$\text{s.t.} \quad X \in \arg\min_{Y\in\mathcal{X}_c} \Phi_c(Y;\alpha,\beta). \tag{45}$$

The admissible coefficients are restricted to the box set

$$\mathcal{H} := \left\{(\alpha,\beta) : \underline{\alpha} \le \alpha^{od} \le \overline{\alpha},\ \underline{\beta} \le \beta^{od} \le \overline{\beta}, \quad \forall (o,d) \in \mathcal{K}\right\}. \tag{46}$$

The purpose of these bounds is to keep the calibration process inside an economically meaningful range and to prevent the market coefficients from drifting toward extreme values during the iterative updates.

The lower-level optimality condition in (45) is implicit. Introducing the lower-level value function from (22), one can rewrite bilevel feasibility as

$$\Phi_c(X;\alpha,\beta) - v(\alpha,\beta) = 0. \tag{47}$$

To handle this condition algorithmically, we use a value-function penalty method and solve problems of the form

$$\Psi_\gamma(X,\alpha,\beta) := -\Pi(X) + \gamma\Big(\Phi_c(X;\alpha,\beta) - v(\alpha,\beta)\Big), \tag{48}$$

with $\gamma > 0$. The penalized objective balances two goals: it seeks designs with high profit, but it also penalizes designs that are far from being lower-level optimal for the current coefficients. This is the device that couples both optimization layers.

The theoretical rationale follows recent value-function-based penalty algorithms for bilevel optimization [7], [8], [10], [12]. Under the regularity assumptions used in that framework, one interprets the penalized problems as approximating the bilevel problem in the sense that stationary points of the penalized sequence approach neighborhoods of locally optimal or bilevel-stationary points when the lower-level value is evaluated accurately enough and the penalty is chosen appropriately. In the present paper we adopt that viewpoint as the formal justification of the algorithmic scheme, while keeping the focus on the practical quality of the resulting network designs.

D. Reweighted ℓ_1 Approximation of Fixed Costs

The original airline design problem contains fixed costs associated with activating airports and hubs. Those costs are naturally related to an ℓ_0-type activation mechanism: once a facility is used, a discrete cost must be paid. Such effects are difficult to handle directly in a smooth continuous model.

To circumvent that difficulty, we approximate fixed costs through a reweighted ℓ_1 procedure derived from the logarithmic surrogate

$$\kappa \log\left(1 + \frac{z}{\varepsilon}\right). \tag{49}$$

This function is much closer to an activation profile than a plain linear penalty. It varies sharply near the origin, thereby mimicking the fact that turning on a previously inactive variable should be expensive, but it becomes flatter as the variable grows. Majorizing (49) at the previous iterate $z^{(k-1)}$ yields the weighted linear term

$$\kappa \frac{z}{z^{(k-1)} + \varepsilon}, \tag{50}$$

which is the classical reweighted ℓ_1 update. Small previous values generate large weights and encourage sparsity, while large previous values are penalized more moderately.

In the proposed model this mechanism is applied to airport and hub capacities and enters the continuous budget restriction (41). As the outer iteration proceeds, the reference capacities are refreshed and the budget surrogate becomes a better approximation of the underlying fixed-cost design logic. This is the component that allows the continuous model to remain sparse and, at the same time, to respect the budget restriction with increasing fidelity.

With the penalty reformulation and the reweighted fixed-cost approximation in place, the full iterative procedure can now be stated explicitly. The next section presents the implemented solution algorithm and explains how the two optimization solves and the outer sparsity updates are coordinated in practice.

V. Solution Algorithm

After the penalty reformulation and the reweighted ℓ_1 approximation have been introduced, the overall solution method can be written explicitly. This section describes how the proposed scheme is implemented in practice. The description follows the actual workflow of `8node_spain/generate_data.py`, together with the two GAMS models `cvx-ll.gms` and `cvx-sl.gms`. The algorithm contains two nested loops: an outer loop that updates the sparse approximation of fixed costs, and an inner loop that

Algorithm 1: Implemented bilevel-sparsity scheme

Input: Initial market coefficients α, β, initial reference capacities $\bar{A}, \bar{S}, \bar{S}^h$, step sizes μ_α, μ_β, penalty parameter γ, numbers of outer and inner iterations.

for $k = 1, \dots, K_{\text{out}}$ **do**
- write the current reference capacities $\bar{A}, \bar{S}, \bar{S}^h$ to the data files used by the GAMS models;
- **for** $t = 1, \dots, K_{\text{in}}$ **do**
  - solve the continuous design model and recover capacities, market shares, and routing variables;
  - evaluate the market-wise terms associated with the lower-level value function;
  - update auxiliary bounds for feasible demand capture;
  - solve the penalized model and recover the corresponding design;
  - compute market-wise corrections for α and β;
  - project the updated coefficients onto their admissible intervals;
  - stop the inner loop early if the relative variation of the objective becomes sufficiently small;
- **end**
- set $\bar{A}, \bar{S}, \bar{S}^h$ equal to the latest continuous-design capacities;

end

Output: Sparse capacities, demand shares, routing pattern, and calibrated market coefficients.

calibrates the market coefficients through repeated interaction between a continuous design solve and a penalized solve.

Initialization

The implementation begins by fixing the number of outer and inner iterations, the step sizes of the market-coefficient updates, and the penalty parameter. The market coefficients are initialized uniformly, and the reference capacities used in the reweighted ℓ_1 approximation are initialized with large positive values. This prevents the first sparsity update from being excessively aggressive and allows the algorithm to explore a broader region of the feasible set.

Outer sparsity iteration

At the beginning of each outer iteration, the reference capacities are written to the text files that define the data of the GAMS models. These references determine the weights appearing in the reweighted fixed-cost budget restriction. Once the inner loop has finished, the algorithm replaces the old references by the capacities produced by the latest continuous-design solve. This update is the majorization-minimization step of the method: it is what makes the logarithmic fixed-cost surrogate evolve from one outer iteration to the next.

Inner bilevel iteration

For fixed reweighting coefficients, the inner loop implements the value-function penalty idea in Algorithm 1. The first solve is performed with `cvx-ll.gms`. Despite its file name, this model plays the role of the continuous inner problem: it returns a design, market shares, and routing variables for the current market coefficients and the current sparsity weights. From that solution, the code computes the market-wise terms that approximate the value function contribution.

After this first solve, the algorithm updates auxiliary bounds on feasible market capture. This step uses the routing pattern induced by the current continuous design in order to avoid implausible logit shares in the subsequent penalized solve.

The second solve is performed with `cvx-sl.gms`. In the present implementation, this model includes the profit term together with the penalized lower-level contribution, and therefore represents the penalized problem associated with the objective Ψ_γ. The design obtained from this second solve is not taken as the new sparsity reference directly. Instead, it is compared with the previous continuous-design solution in order to generate the update direction for the market coefficients.

Market-coefficient update

Once both solves have been computed, the algorithm constructs market-wise corrections for α^{od} and β^{od}. These corrections are based on the discrepancy between the lower-level contribution evaluated at the continuous-design solution and the analogous contribution evaluated at the penalized solution. Intuitively, if a market is underemphasized by the current inner model relative to the profit-oriented penalized solve, its coefficient is reinforced; if it is overemphasized, its coefficient is reduced.

After the raw corrections have been computed, the coefficients are updated by projected steps and clipped to their admissible bounds. The diagonal entries are reset to their nominal values because self-markets do not play any role in the network design. This projection step is part of the algorithmic design, not merely a numerical patch: it guarantees that the learned coefficients remain within a stable and interpretable range.

Transfer between loops

At the end of the inner loop, the algorithm transfers to the outer loop the capacities returned by the continuous-design solve rather than the penalized one. This choice is deliberate. The outer loop is meant to update the sparse approximation of the inner problem itself, so the correct reference is the design that approximates the value function, not the design generated only to compute the penalty correction. If this transfer were performed incorrectly, the reweighted budget surrogate would gradually drift away from the design model that it is intended to approximate and could easily stop respecting the intended budget logic.

Interpretation

The final method should be read as a nested successive-approximation scheme. The outer loop improves the representation of fixed activation costs through reweighted ℓ_1 updates. The inner loop improves the market priorities through a value-function penalty method. The first mechanism enforces sparsity and helps preserve the budget structure, while the second mechanism aligns the continuous design model with the true profit objective of the airline. Their joint action is what makes the approach computationally viable and economically meaningful.

VI. Conclusion

We have presented a continuous bilevel framework for airline hub-and-spoke design in which sparse deployment decisions, logit demand capture, and profit maximization are treated within a unified optimization model. The lower level determines capacities, flows, and market shares under a continuous approximation of fixed costs, while the upper level learns market-specific hyperparameters that steer the design toward more profitable market allocations.

The resulting model preserves the main practical advantages of continuous optimization. In particular, it remains scalable and directly compatible with rich passenger-choice terms. At the same time, the penalty-based bilevel strategy provides a concrete mechanism for coupling the learned market priorities with the true economic objective of the airline. For that reason, the framework is a promising alternative to exact mixed-integer bilevel formulations for large airline network design problems.

The next step is to deepen the numerical section: report the effect of the learned market hyperparameters on the resulting network topology, compare the penalty-based bilevel design against the underlying single-level model, and quantify how the outer sparsity loop affects both profitability and the number of hubs selected.

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